

MIT Art Design and Technology University's
MIT School of Computing, Pune
Department of Computer Science and Engineering
B. Tech Third Year A.Y.2025-26



Machine Learning Lab

Enrolment No: _____ Division: _____

Roll No: _____ Name: _____

Experiment No.: 03

Title: Write a program to implement k-Nearest Neighbour algorithm to classify the iris data set. Print both correct and wrong predictions. Python ML library classes can be used for this problem

Theory:	<i>Introduction</i> The k-Nearest Neighbour (k-NN) is a supervised learning algorithm used for both classification and regression tasks. It is one of the simplest and most intuitive machine learning algorithms. The core idea is to predict the class of a data point based on the majority class of its k nearest neighbors in the feature space. <i>Working of k-NN Algorithm</i> 1. Load the dataset (Iris dataset in this case). 2. Choose the number of neighbors k. 3. Calculate the distance (commonly Euclidean) between the query instance and all training samples. 4. Identify the k-nearest data points to the query instance. 5. Perform majority voting for classification: assign the class which appears most frequently among the k neighbors. 6. Return the predicted class label. <i>About the Iris Dataset</i> • A classic dataset for classification problems. • Contains 150 records of iris flowers from 3 species: <code>setosa</code>, <code>versicolor</code>, and <code>virginica</code>. • Each record has 4 features: 1. Sepal length 2. Sepal width 3. Petal length 4. Petal width
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Distance Metric

- Most commonly used distance metric is **Euclidean Distance**:

$$d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2}$$

Advantages of k-NN

- Simple to understand and easy to implement.
- No training step — it is a **lazy learner**.
- Effective with a small number of input variables.

Disadvantages

- Computationally expensive for large datasets.
- Sensitive to **irrelevant or redundant features**.
- Performance depends heavily on the choice of **k** and the **distance metric**.

Evaluation

After implementing k-NN:

- Compare predicted labels with actual labels.
- Count **correct predictions** and **wrong predictions** to evaluate performance.

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Flowchart: (if any)	
Pseudo code or Program	<pre> # Import required libraries from sklearn.datasets import load_iris from sklearn.model_selection import train_test_split from sklearn.neighbors import KNeighborsClassifier from sklearn.metrics import accuracy_score # Load the Iris dataset iris = load_iris() X = iris.data y = iris.target target_names = iris.target_names # Split the data into training and testing sets (70% training, 30% testing) X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42) # Create KNN classifier (you can change n_neighbors as desired) knn = KNeighborsClassifier(n_neighbors=5) # Train the model knn.fit(X_train, y_train) # Make predictions y_pred = knn.predict(X_test) # Calculate accuracy accuracy = accuracy_score(y_test, y_pred) print(f"\nModel Accuracy: {accuracy * 100:.2f}%\n") # Print both correct and incorrect predictions print("==== Prediction Results ====") for i in range(len(y_test)): actual = target_names[y_test[i]] predicted = target_names[y_pred[i]] if y_test[i] == y_pred[i]: print(f" Correct: Predicted = {predicted}, Actual = {actual}") else: print(f" Wrong: Predicted = {predicted}, Actual = {actual}") </pre>

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Outputs	
Exercise for Practice	
Q1.	Explain how distance is calculated between points in the k-NN algorithm. Why is Euclidean distance commonly used?
Ans:	
Q2.	What preprocessing steps are important before applying k-NN to a dataset, and why??
Ans	
Q3	Suppose your k-NN classifier gave an accuracy of 92% on the test set. Suggest at least two ways to possibly improve the model performance?
Ans	

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Q4.	What are the main advantages and disadvantages of using the k-NN algorithm for classification tasks?
Ans	
Conclusion	Thus, we have successfully implemented Knn algorithm on a dataset
Date of Submission:	
Marks out of 10	Name and Sign of Subject Teacher
Remark:	